



Jacob González Isa

PhD student at the University of Cambridge

Department of Plant Sciences & King's College Cambridge

MSc in Computational Biology | BSc (Hons) in Biochemistry

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Personal Bio

As I began using bioinformatic tools in my early computational biology lessons, I quickly recognized the importance of this discipline in the field of Life Sciences.

By harnessing the power of Computational Biology and Synthetic Biology, I am driven to unravel the complexities of bio-sciences and unlock novel insights that can make an impact in the areas I am passionate about.

Plants are my plans for the future!

Research Experience

From October
2024

PhD student in Plant Sciences (Bioinformatics) | Postgraduate “la Caixa” Fellow

Department of Plant Science, University of Cambridge, Cambridge (UK)

- Funded by the competitive “la Caixa” Foundation Postgraduate Abroad Fellowship, awarded from the hands of the King of Spain Felipe VI.
- Working in the group of **Genetic and Epigenetic Inheritance in Plants** (Prof. Henderson) (<https://www.plantsci.cam.ac.uk/research/groups/genetic-and-epigenetic-inheritance-plants>)
 - Primarily work as bioinformatician (pipeline development, design of phylogenomic studies, long-read analysis, comparative genomics across species, etc.). Linux shell, SLURM, Python and R.
- Supervisor of the course “Mathematical and Computational Biology” (Part II/second year)
 - More than 20 hours of teaching, plus revisions of exercise sheets.
 - Project marker (more than 30 hours of marking projects of the course).
- 1st year graduate representative
- College activities (King’s College Cambridge):
 - 1-year Entrepreneurship Lab offered by King’s College and Social Venture Residential Programme
 - Entrepreneurship Lab Social Venture Residential
 - Tennis team of King’s College

January 2024 -
August 2024

MSc thesis - Research Intern

CBGP (Centre of Plant Biotechnology and Genomics), Madrid (Spain)

- Working in phylogenomics and comparative genomics within the research group of Jaime Huertas-Cepas (<http://compgenomics.org/team/>)
 - Advanced bash, R and Python integrated in High Performance Computing (HPC) with SLURM.
 - ETE toolkit for phylogenomic analysis.
- MSc thesis: “Detection of metagenomic functional enzyme variants of biotechnological and ecological interest through phylogenomics” (grade: 10/10)

July 2023 -
September 2023

Erasmus+ Research Intern

ISTA (Institute of Science and Technology Austria), Vienna (Austria)

- Funded by Erasmus+ Internship Scheme
- Working in Prof. Friml group, international reference in the auxin signalling field (<https://ist.ac.at/en/research/friml-group/>)
 - CRISPR/Cas9-mediated targeted mutagenesis in *Marchantia polymorpha*
 - Cloning (Gateway, Gibson assembly, ...), Agrobacterium-mediated transformation
 - Optimization of protoplast isolation of the microalga *Drapalnadia*

October 2022 -
June 2023

JAIE Intro CSIC (National Spanish Research Council) Research Intern

IRNAS (Institute of Natural Resources and Agrobiology of Seville), Seville (Spain)

- Working in the group of Ionic and Water Regulation in plants for eight months (<https://www.irnas.csic.es/regulacion-ionica-e-hidrica-en-plantas-rih/>)
 - Set up of plant cultures in hydroponics and soil
 - DNA extraction and genotyping
 - RNA-seq
 - Plant material collection, preservation in liquid nitrogen and lyophilization
 - RT-PCR
- Funded by the competitive **JAIE-Intro Scholarship**

October 2022 -
June 2023

BSc Thesis - Research intern

Department of AI and Computer Science, University of Seville, Seville (Spain)

IBVF (Institute of Plant Biochemistry and Photosynthesis), Seville (Spain)

- Funded by the competitive **Scholarship by the Ministry of Education of Spain for Collaboration at University Departments**
- Working in the group of Systems Biology and Biotech in Microalgae (https://greennetwork.us.es/systems_biology_biotechnology_microalgae/)
 - Microalgae culture
 - Optimization of ChIP-seq protocol
 - Western blotting
 - Bioinformatic analysis (cistromics and transcriptomics)
 - R, Python and bash scripting.
- BSc thesis: "Characterization of the epigenetic modification H3K27me3 in *Klebsormidium nitens*" (grade: 9.5/10)

July 2022 -
September 2022

Amgen Scholars Programme - Fully funded internship

Department of Plant Science, University of Cambridge, Cambridge (UK)

- One of the most competitive Summer Research Programmes in Europe. Funded by **Amgen Foundation and coordinated by Harvard University**
- Working in the group of Genetic and Epigenetic Inheritance in Plants (Prof. Henderson) (<https://www.plantsci.cam.ac.uk/research/groups/genetic-and-epigenetic-inheritance-plants>)
 - R and bash programming
 - Sequence alignments and phylogenetic analysis

Publications

Piotr Włodzimierz*, Estela Perez-Roman*, Michael Hong , Meng Zhang, Ludmila Oliveira, **Jacob Gonzalez-Isa**, Katharine Jenike, Robin Burns, Chenxi Zhou, Sagarika Koppera, Mia Becker, Nicola Gorringer, Petr Novak, Yennifer Mata-Sucre, Ioannis Kontoyiannis, Mark Blaxter, André Marques, Jiri Macas, Alexandros Bousios, Richard Durbin, Ian Henderson. Cyclical evolution of centromere architecture across 193 eukaryote species. *Submitted* (2025)

Media Outreach and Mentions

- [Jacob González, the young Andalusian who is shining at Cambridge with his scientific career](#)

TV interview for "La tarde, aquí y ahora" (biggest audience in CanalSur channel). August 2025

- [A scientist from Seville at the University of Cambridge: 'There are still prejudices about Andalusians, but they are gradually disappearing.'](#)

ABC newspaper (with the biggest audience in Andalusia) interviewed me and published a series of four articles. TikTok and Instagram shorts were also released, which accumulated around 1 million views. July 2025

- [They study in Cambridge](#)

Radio interview by Ainhoa Caballero for the radio program "Spaniards Abroad".

RNE (Radio Nacional Española) is Spain's national public radio broadcaster. July 2025

- [A student from Seville receives a grant from the 'la Caixa' Foundation to study a postgraduate course abroad](#)

"Aljarafe Noticias" Newspaper, June 2025

- [Bioinformatics, Spain and I](#)

Blog and website of the Open Bioinformatics Foundation (OBF)